

Session 9: ArraysTopics:1D & 2D Arrays, Array Traversal.Example: Record 7-day screen time & calculate average.

Tasks

1.

Declare a 1D array called dailySteps with 7 elements to store your step count for each day of the week, assign sample values, and print each value using a loop.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int dailySteps[7] = {5000, 7200, 6500, 8000, 4500, 9000, 7500};
```

```
    int i;
```

```
    for(i = 0; i < 7; i++)
```

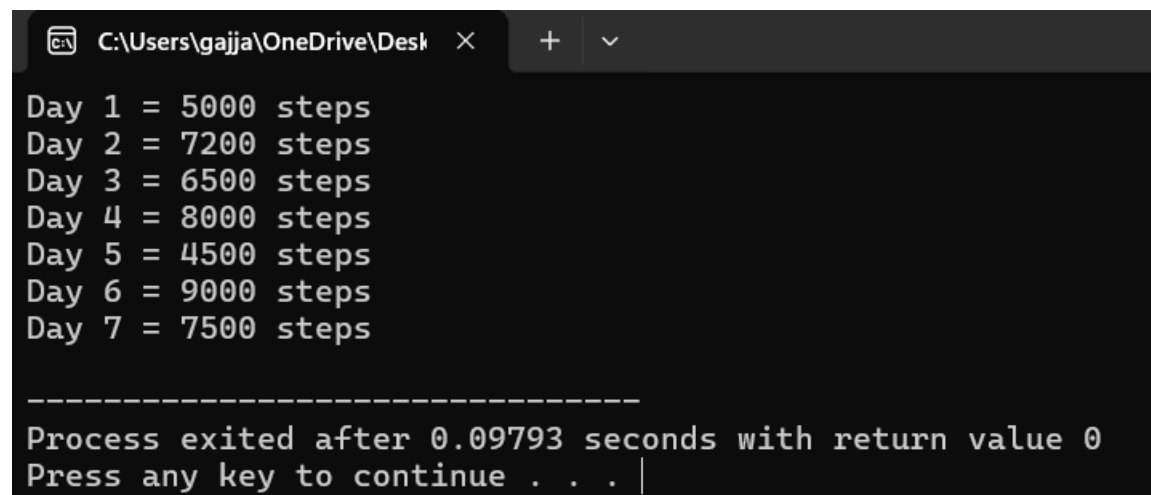
```
    {
```

```
        printf("Day %d = %d steps\n", i + 1, dailySteps[i]);
```

```
    }
```

```
    return 0;
```

```
}
```



```
C:\Users\gajja\OneDrive\Desktop >
Day 1 = 5000 steps
Day 2 = 7200 steps
Day 3 = 6500 steps
Day 4 = 8000 steps
Day 5 = 4500 steps
Day 6 = 9000 steps
Day 7 = 7500 steps

-----
Process exited after 0.09793 seconds with return value 0
Press any key to continue . . . |
```

2.

Create a 2D array called `playlistRatings` to store ratings for 3 Spotify playlists over 5 days (rows = playlists, columns = days). Fill it with sample numbers and print the ratings for the second playlist.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int playlistRatings[3][5] =
```

```
    {
```

```
        {4, 5, 3, 4, 5},
```

```
        {5, 4, 5, 5, 4},
```

```
        {3, 4, 4, 5, 3}
```

```
    };
```

```
    int i;
```

```
    printf("Ratings of second playlist:\n");
```

```
    for(i = 0; i < 5; i++)
```

```
    {
```

```
        printf("Day %d = %d\n", i + 1, playlistRatings[1][i]);
```

```
    }
```

```
    return 0;
```

```
}
```

```
C:\Users\gajja\OneDrive\Desk × + v
Ratings of second playlist:
Day 1 = 5
Day 2 = 4
Day 3 = 5
Day 4 = 5
Day 5 = 4

-----
Process exited after 0.1087 seconds with return value 0
Press any key to continue . . . |
```

3.

Build a function that takes a 1D array of 7 integers representing your daily Zomato order amounts and calculates the average spend for the week.

Hint: Use a loop to sum the values, then divide by the array length.

```
#include <stdio.h>
```

```
float calculateAverage(int orders[], int size)
```

```
{
```

```
    int sum = 0;
```

```
    int i;
```

```
    for(i = 0; i < size; i++)
```

```
    {
```

```
        sum = sum + orders[i];
```

```
    }
```

```
    return (float)sum / size;
```

```
}
```

```
int main()
```

```

{
    int orders[7] = {200, 350, 150, 400, 250, 300, 350};

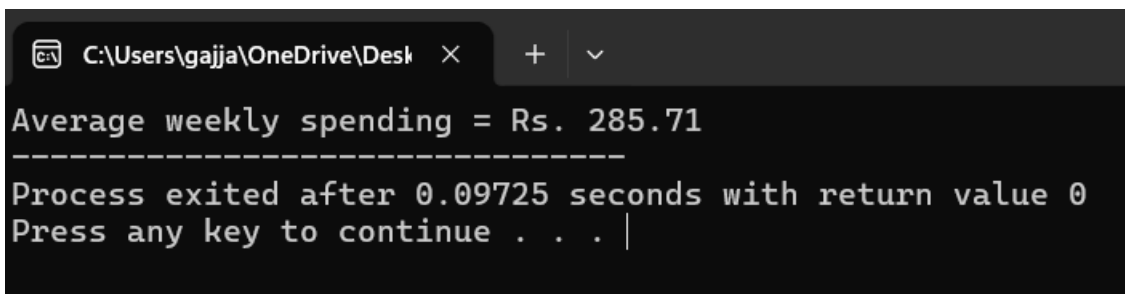
    float average;

    average = calculateAverage(orders, 7);

    printf("Average weekly spending = Rs. %.2f", average);

    return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop >
Average weekly spending = Rs. 285.71
-----
Process exited after 0.09725 seconds with return value 0
Press any key to continue . . . |

```

4.

Given a 2D array called cricketScores where each row represents an IPL match and columns represent runs scored by each team, write code to print the highest score from each match.

```
#include <stdio.h>
```

```

int main()
{
    int cricketScores[5][2] =
    {
        {185, 172},
        {210, 198},
        {156, 160},

```

```
{202, 190},  
{175, 180}  
};
```

```
int i;
```

```
int highest;
```

```
for(i = 0; i < 5; i++)
```

```
{
```

```
    if(cricketScores[i][0] > cricketScores[i][1])
```

```
    {
```

```
        highest = cricketScores[i][0];
```

```
    }
```

```
    else
```

```
    {
```

```
        highest = cricketScores[i][1];
```

```
    }
```

```
    printf("Match %d highest score = %d\n", i + 1, highest);
```

```
}
```

```
return 0;
```

```
}
```

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```
Match 1 highest score = 185  
Match 2 highest score = 210  
Match 3 highest score = 160  
Match 4 highest score = 202  
Match 5 highest score = 180
```

```
-----  
Process exited after 0.09329 seconds with return value 0  
Press any key to continue . . . |
```